

Nafis Imtiaj Hossain

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RESEARCH INTEREST

- Estuarine ecosystem health, phytoplankton dynamics, biogeochemical cycles.
- GIS and Remote Sensing for marine, coastal landscape and vulnerability assessments
- Multivariate statistical techniques and regression models using R/Python
- Climate & vulnerability indices to improve community risk management

EDUCATION

Degree	Major	Institution	CGPA (/4)	Duration
M.Sc.	Water Resources Development	Bangladesh University of Engineering and Technology (BUET)	3.08 (72%)	2023 - 2026
B.Sc.	Oceanography	Bangladesh Maritime University	3.45 (69%)	2017 - 2022

- ✓ **Master's thesis:** *Comparative analysis of risks, losses and damages from salinity intrusion in Coastal Bangladesh.*
 - Analyzed climate risk, exposure, sensitivity and adaptability using ETCCDI indices and LVI, CRM models.
 - Evaluated loss and damage using multivariate dimensionality reduction techniques i.e. PCA, CCA, RDA, K-means cluster including polynomial & penalized regression models.
- ✓ **Bachelor's thesis:** *Environmental Drivers of Phytoplankton Community in the Sundarbans Mangrove, Bangladesh.*
 - Assessed phytoplankton community structure using diversity, richness, evenness & distribution indices.
 - Combined nutrients and heavy metal indices feed into a phytoplankton quality index (PQI).

TECHNICAL SKILLS

Oceanographic Instruments	CTD, Van Veen Sediment Grab, Multimeter, Algal Torch, Niskin Water Sampler, ADCP, GPS (RTK-Trimble, Garmin Handheld), Total Stations.		
Laboratory Instruments	AAS(Shimadzu), UV Spectrophotometer, Autoclave, Microwave Digestion, Multi-chamber Thermo Incubator, Autoanalyzer.		
Statistical tools	SPSS, PAST, OriginPro	Coding	Python, Rstudio
Mapping	ArcGIS Pro, ArcMap, GEE, QGIS	Qualitative Analysis	NVivo, KoboToolbox
Referencing	Mendeley Desktop, Zotero	Productivity	MS Office

PROFESSIONAL EXPERIENCES

- United Nations Development Programme - UNDP, Bangladesh** Sep 2025 – Jun 2026
Research Assistant 10 months (full-time)
- ✓ **Assisted the Governance of Climate Change Finance -GCCF & Climate Finance Network -CFN** projects in Asia-Pacific
 - Supported the development of Climate Damage Function (CDF) incorporating CVI, CRI & asset loss data.
 - Supported the Ministry of Planning in integrating climate change appraisal criteria for development projects.
- Center for Participatory Research and Development - CPRD** Jun 2025 - Aug 2025
Research Officer 3 months (full-time)
- ✓ **Strengthened the Climate Resilient Green Accountability for WASH project with WaterAid** ([Link](#)),
 - Mapped climate finance flows in urban and coastal vulnerable areas of Bangladesh ([Link](#)).
- Oxfam in Bangladesh** Mar 2024 - Mar 2025
Young Fellow 1 year (Hybrid)
- ✓ **1 year fellowship under Oxfam's fellowship programme targeting youth in Bangladesh** ([Link](#))
 - Mapped flood and livelihood vulnerability using GIS-RS & CRM factors in drought prone areas.
 - Blended hydrological data feeding into an IWRM model using MCDA tool.
- International Centre for Climate Change and Development - ICCCAD** Jun 2024 - May 2025
Research Assistant ([Link](#)) 1 year (full-time)
- ✓ **Backed the Strengthening Social Resilience Program** ([ADB](#)) & **Addressing loss & damage of LDCs projects** ([CASA](#))
 - Assessed readiness, GEDSI inclusiveness of [EGPP](#) & reviewed barriers to active social adaptation programs.

- Mapped hazard, vulnerability and loss & damage of migrants & non-migrants in coastal slums.

Intern

Jul - Dec 2023, 6 months (full-time)

- Prepared Hazard Calendar, Climate Vulnerability Assessment (CCVA) and Local Climate Adaptation Plan in coastal towns under the [CRMFT through LLA](#) & [CAP-RES](#) projects.

Nirmol Bangladesh

Jul 2022 - Jun 2023

Research Coordinator

1 year (full-time)

- ✓ **Supported the Ocean Literacy for Climate Resilience for vulnerable gendered groups (Korea SHE)** ([Link](#))
 - Trained school students on marine skills ensuring gender inclusive sustainable ocean practice ([full report](#)).

Bangladesh Maritime University

Oct 2019 - June 2022

Research Assistant

2.7 years (part-time)

- ✓ **Facilitated the Fisheries diversity & environmental variables in Sundarbans mangrove waterways project (UGC)**
 - Guided research teams to collect data from the Bay of Bengal (BoB) coast i.e. Sundarbans, Saint Martin's Island.
 - Conducted microscopic identification, and enumeration of ecological components to assess ecosystem health.
- ✓ **Facilitated the Natural & Anthropogenic Threats to Sustainable Management of Nijhum Dwip MPA (UGC)**
 - Conducted ecotoxicity risk assessments using standard pollution indices of water and sediment samples.
 - Performed statistical analysis with SPSS, R, and Python to evaluate nutrient, heavy metal & grain size data.

PUBLICATIONS

Journal Articles

1. **Hossain, N. I.**, Rahman, S. M. M., Hossain, Md. A., Bin Hafiz, K. & Masud, M. (2025). Decoding phytoplankton quality in the waterways of the Sundarbans mangrove forest of Bangladesh. *Marine Pollution Bulletin*, 117669. <https://doi.org/10.1016/j.marpolbul.2025.117669>
2. Hafiz, K. B., Khan, A. A., Rahman, S. M. M., **Hossain, N. I.**, M. A., Maksud, M. A., Alam, M. N. E., Rahman, M. S. & Choudhury, T. R. (2025). Assessment of heavy metal concentrations in seabed sediments of the coral ecosystem in Bangladesh: An in-depth analysis of ecotoxicity and ecological health. *International Journal of Sediment Research*, <https://doi.org/10.1016/j.ijsrc.2025.11.001>
3. Sowrav, S. F. F., Hossain, Md. A., Hafiz, K. B., **Hossain, N. I.** (2023). Multi-dimensional approach for an environmental health assessment of a deltaic mangrove ecosystem, Sundarbans. *Environmental Advances*, 12, 100377. <https://doi.org/10.1016/j.envadv.2023.100377>

Book Chapter

1. **Hossain, N. I.**, Meem, T. Z., Khatun, S., Ritu, S. M., Murshed, S. B., A Comprehensive Integrated Water Resources Management (IWRM) Plan for Dharla River: Identifying Human Impact on Water Resources (2024). *Proceedings of the 7th International Conference on Advances in Civil Engineering (ICACE2024)*, *Sustainable Design and Construction: Selected Papers from ICACE2024*, V1. Springer Singapore.

Conference & Posters

1. Das, M. K., Hossain, M. A., Sowrav, S. F. F., **Hossain, N. I.**, Hossain, M. S. (2025). Hydrometeorological Analysis of the Unprecedented Flood in Eastern Bangladesh *10th International Conference on Water and Flood Management, IWFM, BUET*.
2. **Hossain, N. I.**, Meem, T. Z., Khatun (2024). Evaluation of the Impact of Dharla River on Agricultural Sector and Developing a Management Plan for the Catchment Area Through Integrated Water Resources Management (IWRM). *Bangladesh University of Professionals Research Society (BUPRS)*, Dhaka, Bangladesh ([poster link](#))
3. **Hossain, N. I.** (2024). Beyond the Banks: Comparing Flood-Induced Livelihood Vulnerability Between Char Land & Inland Communities Using GIS-RS Based Approach. *2nd South Asian Conference, Centre for People and Environ (CPE)*, Dhaka ([Poster](#)).
4. Rahman, S. M. M., Sowrav, S. F. F., **Hossain, N. I.**, (2025). Integrated Assessment of Pollution Dynamics and Ecological Impacts in the Rupsha-Pashur River System and the Sundarbans Mangrove Ecosystem. *International Symposium on Marine Litter & Coastal Ecosystem Nexus*, Khulna University, Bangladesh.

Policy Briefs & Knowledge Products

1. Policy workshop report: Equity and Co-management of Baikka Beel Permanent Wetland Sanctuary (2023), **Co-authored** [USAID Ecosystems/Protibesh Activity](#), ICCCAD, Dhaka, Bangladesh
2. 30th Conference of the Parties: Articulating CSOs Position Together (2025). Co-author, *Climate Justice Alliance-Bangladesh housed at Center for Participatory Research and Development-CPRD* in Dhaka, Bangladesh
3. Saleemul Huq Action Research Exchange - **SHARE Hub** at COP 29 by compiling climate policy related thematic sessions and youth-led climate actions, [ICCCAD, Dhaka, Bangladesh](#) (2024).
4. Empowering Local NGOs: Bridging Capacity Gaps for CC Action in Kurigram (2023). Co-authored, *Climate Tribune* (2023).
5. Learning Report synthesizing key conference interventions to drive evidence-based adaptation policies ([Link](#)).
6. South Asia Resilience Hub, detailing priorities & climate challenges to inform the Race to Resilience agenda ([Link](#)).
7. Module on Locally led adaptation and building community resilience ([Link](#)).

AWARDS & ACHIEVEMENTS:

Runner-up [May 2024] National Research Contest 3.0, BUP Research Society ([Link](#)).

Best Trainee [Sep 2021] in certificate course on Data Science and Machine Learning with Python under NOAMI ([Link](#)).
VC's Scholarship [2019 & 2020] for achieving SGPA > 3.75 during B.Sc. studies at Maritime University, Bangladesh.

WORKSHOPS & TRAINING ACTIVITIES

- Data sorting, visualization & storytelling using Power BI: **Data Analytics with Power BI**, IIL ([Link](#)) 7 day | 2024
- Application of dimensionality reduction techniques: **National Data Science Bootcamp**, BDSS ([Link](#)) 1 day | 2023
- **Data Science & Machine Learning with Python**, NOAMI ([Link](#)) 1.5 months | 2024
- Basic statistical analysis of biological data: **Bioblitz: Biodiversity Conservation**, GYBN 3 day | 2020
- Weather-based risk insurance products: **Training Course: Climate Risk Insurance**, CPRBD, DU ([Link](#)) 7 days | 2024
- Just Energy Transition, RMG sector: **Pathways to JET for a Sustainable Future**, Oxfam 3 days | 2025

REFEREES

1. Dr. SM Mustafizur Rahman – Assistant Professor,
Bangladesh Maritime University

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2. Dr. A.K.M. Saiful Islam – Professor, IWFM
Bangladesh University of Engineering & Technology

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